MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FIRST SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)

MG3DSEECT205 - Microcontroller Programming

Duration: 2 hours Maximum Marks: 35

Practical Examination – Model Question Paper

Instructions:

- **Demonstrate** the given task. (Any one of the tasks listed below)
- **Viva session** (Minimum 5 questions from the Practical Module)
- **Submit Certified Lab Report**

Evaluation Criteria:

- **Viva**: 7 marks
- **Lab Report**: 8 marks
- **Demonstration**: 20 marks

Experiments: (Choose any one)

1. Write a Python code to blink an LED connected to a Raspberry Pi.
2. Write a Python code to turn an LED on/off when a push button is pressed(using Raspberry Pi).
3. Create a Python program to blink three LEDs in a sequence.
4. Design a traffic light controller using Raspberry Pi and LEDs.
5. Write a Python program to turn on an LED when motion is detected using a motion sensor.
6. Create an Intruder Alert System using a motion or temperature sensor and a buzzer with Raspberry Pi
7. Write a Python program to perform AND, OR, and NOT operations on two input values.
8. Write a Python program to toggle the bits of a given number using bitwise operators.
   a. 5
   b. 10
9. Write a Python program to calculate the sum of a list of numbers.
   a. [3, 5, 2, 7, 1, 9]
   b. [10, 20, 30, 40, 50]
10. Write a Python program to concatenate two strings and display the result.

a. Hello World

b. Python is awesome

11. Write a Python program to find the largest and smallest numbers in a given list of numbers.

    a. [12, 45, 7, 23, 56, 89, 34]

    b. [90, 10, 30, 50, 70, 20, 40]

12. Write a Python program to display all even numbers from 1 to 10.

13. Write a Python program to take a number as input and display a string along with the number.

    For example, number = 42

    string = "Your favourite number is: "